Algorithm to Compare Maps of a Continuous Variable at Multiple Resolutions

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Abstract

Context: Mean Absolute Deviation (MAD) measures the average of the absolute differences between X and Y. MAD is applicable for situations where X and Y show the same phenomenon. Applications can include differences between time points, differences between predicted versus observed values, and differences attributable to the technique of observation.

Objectives: This paper explains MAD's three components: quantity, stratum, and element. This paper illustrates the components by examining maps of global sea surface temperature for two comparisons: August 1982 versus February 1982 and February 2010 versus February 1982.

Methods: A scatter plot of the observations with respect to the Y=X line reveals how X and Y disagree in terms of quantity and allocation. The inclusion of stratification allows for the extraction of components of MAD that separate and quantify different types of deviation. This analysis is performed at multiple resolutions to show the effects of changing the spatial resolution.

Results: The quantity component of MAD is the largest component for the change in sea surface temperature from February 1982 to February 2010 due to warming in both the northern and southern hemispheres. The stratum component of MAD is the largest component for the change in sea surface temperature from February 1982 to August 1982 due to the warming in the northern hemisphere and cooling in the southern hemisphere. The element component shrinks as spatial resolutions become coarser.

Conclusions: MAD has desirable properties. MAD is unbounded, symmetric, and is expressed in the same units as the deviations that it measures. MAD is useful for comparing maps because MAD at multiple spatial resolutions measures how far positive deviations are from negative deviations in space. MAD should be computed, reported and interpreted when comparing images of the same continuous variable.

Keywords

Error, change detection, mean absolute deviation, sea surface temperature

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1 Introduction

1.1 Research purpose

This paper presents a measure of deviation that quantifies differences between maps X and Y of the same continuous variable. The Mean Absolute Deviation (MAD) measures how far on average points of a scatter plot are from the Y=X line. Deviations are the observations of map Y minus the observations of map X. This measure has been shown previously to provide easily interpretable and meaningful information in its results (Pontius et al. 2008; Willmott and Matsuura 2005; 2006).

MAD can be used in a wide range of fields, including climatology, land change, and remote sensing. This measure could be used to characterize differences between time points or used to validate model or sensor performance. Applications could include performing MAD analysis on predicted versus observed values, and measuring differences attributable to the method of data collection. To compare two predictive models, MAD should be run twice, with a constant reference map as X and changing images Y. In applications, it will be necessary that both maps present the same continuous variable. For example, one could run this type of analysis on two precipitation maps, but not on maps of precipitation versus temperature.

1.2 Literature review

In an assessment of measures of average error, Willmott and Matsuura (2006), concluded that the mean absolute error should be reported for applications concerning

model evaluation and cross-site comparisons of maps that share the same variable. Willmott et al. (2009) offer valuable insight to available statistical measures for geographic interpolation models and outlines the problems with various error statistics such as root mean squared deviation. The mean square error and root mean square error are functions of the average error and the variation among absolute value of the errors. Squared errors do not describe average error alone. Willmott and Matsuura (2005) report the characteristics that lend the mean absolute error to be an unambiguous measure for model-performance.

Pontius et al. (2008) explains a logic of analysis using a proposed map X as reference data which can be compared to a map Y. Maps can differ in terms of quantity and allocation. In contrast to the root mean square error, the approach described in Pontius et al. (2008) relies on simpler mathematics, using the mean absolute error. Pontius (2010) provides an example of extracting quantity and allocation disagreement for categorical variables. Pontius et al (2008) also addresses the need to measure at multiple resolutions. How measurements change as a function of resolution can be more important and revealing than results given at a single resolution. Measuring at multiple resolutions accounts for the influence of data formatting. Willmott and Matsuura (2006) warn that the use of inappropriate measures can lead to incorrect interpretations of performance. Heeding this warning, this paper will encourage the use of MAD as an unambiguous measure.

2 Methods

2.1 Data

Figure 1 shows all the inputs used in the analysis. There are three maps of sea surface temperature in °C at different time points: February 1982, August 1982, and February 2010, as well as a stratum map that identifies the Northern and Southern hemisphere. These maps are displayed using the equal area projection, Molleweide Pacific Centered. The spatial resolution of each pixel side is 100 km. NOAA Optimum Interpolation Sea Surface Temperature V2 data are provided by the NOAA/OAR/ESRL PSD, Boulder, Colorado, USA, from their Web site at http://www.esrl.noaa.gov/psd/.

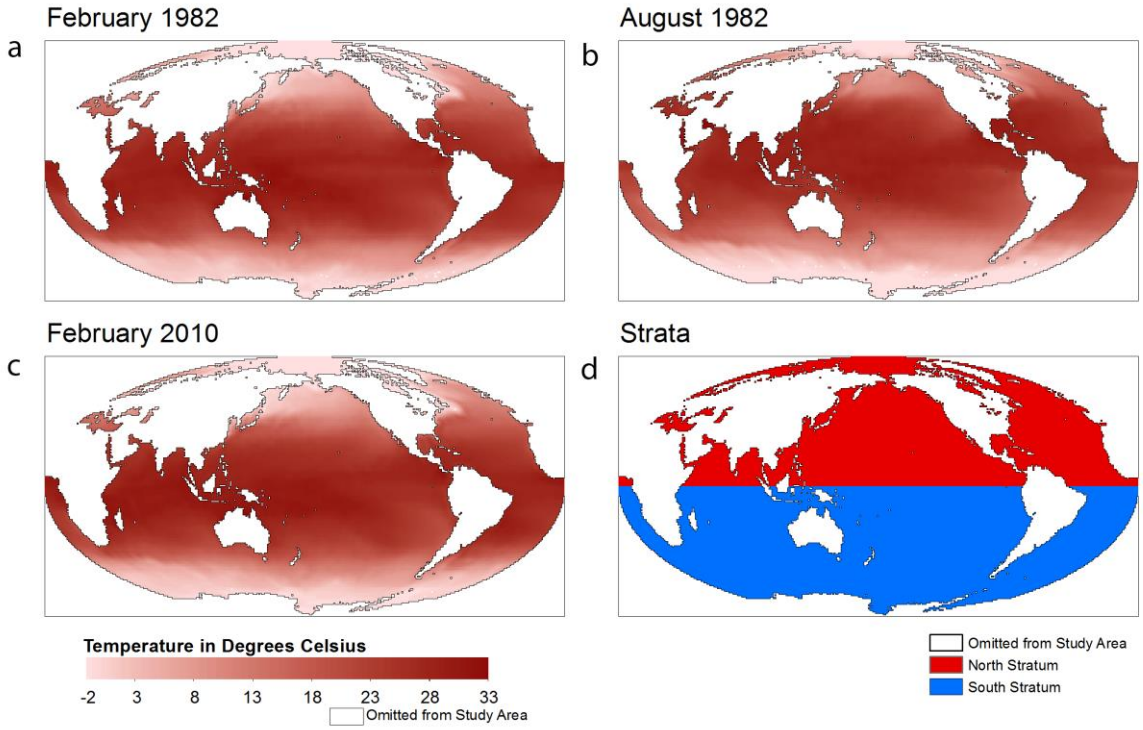

Figure 1. Input Raster maps at 100-km resolution showing Sea Surface Temperature at three time points and two strata.

Figure 2 shows deviations across six months within the same year. Deviations for each pixel are plotted as time point Y versus time point X with respect to the Y=X line. A pixel displayed in red on the map indicates warming, while a pixel displayed in blue on the map indicates cooling. If the temperature is the same at both time points for a pixel, it is yellow in the map, and its corresponding point on the scatter plot is on the Y=X line. In Figure 2, points that are above the Y=X line tend to be from the northern hemisphere. Points that are below the Y=X line tend to be from the southern hemisphere. The scatter plot in figure 2 reflects spatial reallocation of energy from the southern hemisphere to the northern hemisphere due to changing seasons.

Figure 3 shows deviations for the same month across different years. The deviation map shows scattered areas of warming and cooling in both strata. Points on the scatter plot are closer to the Y=X line and there is no distinct spatial pattern of temperature deviation values between the northern and southern hemispheres. Points in blue, as well as points in red, are seen above and below the Y=X line.

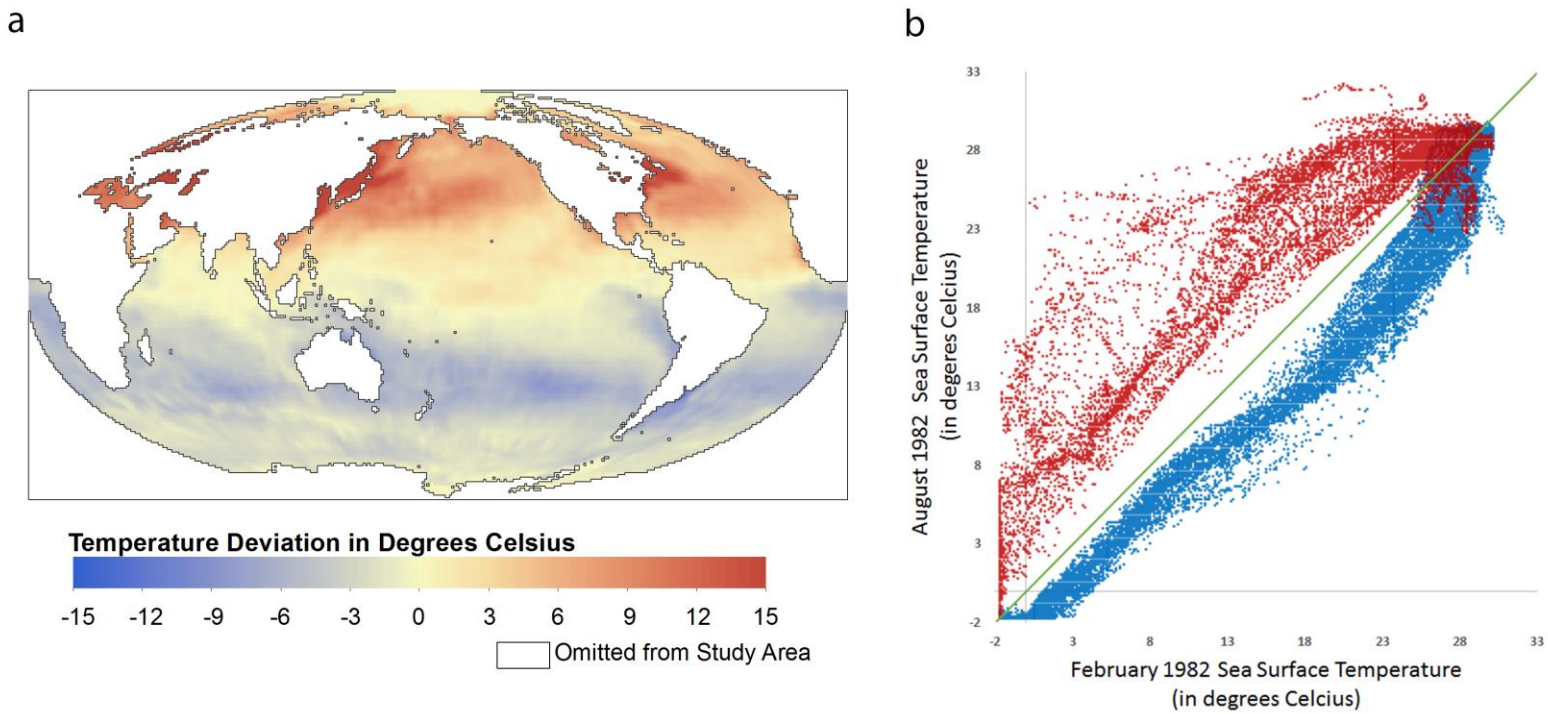

Figure 2. Map and scatterplot of temperature deviations for August 1982 minus February 1982. Each pixel on the map is represented as a point on the scatter plot to the right, with pixels in the North Stratum displayed in red and pixels in the South Stratum displayed in blue. The Y=X line in green shows where temperature values are the same for both time points.

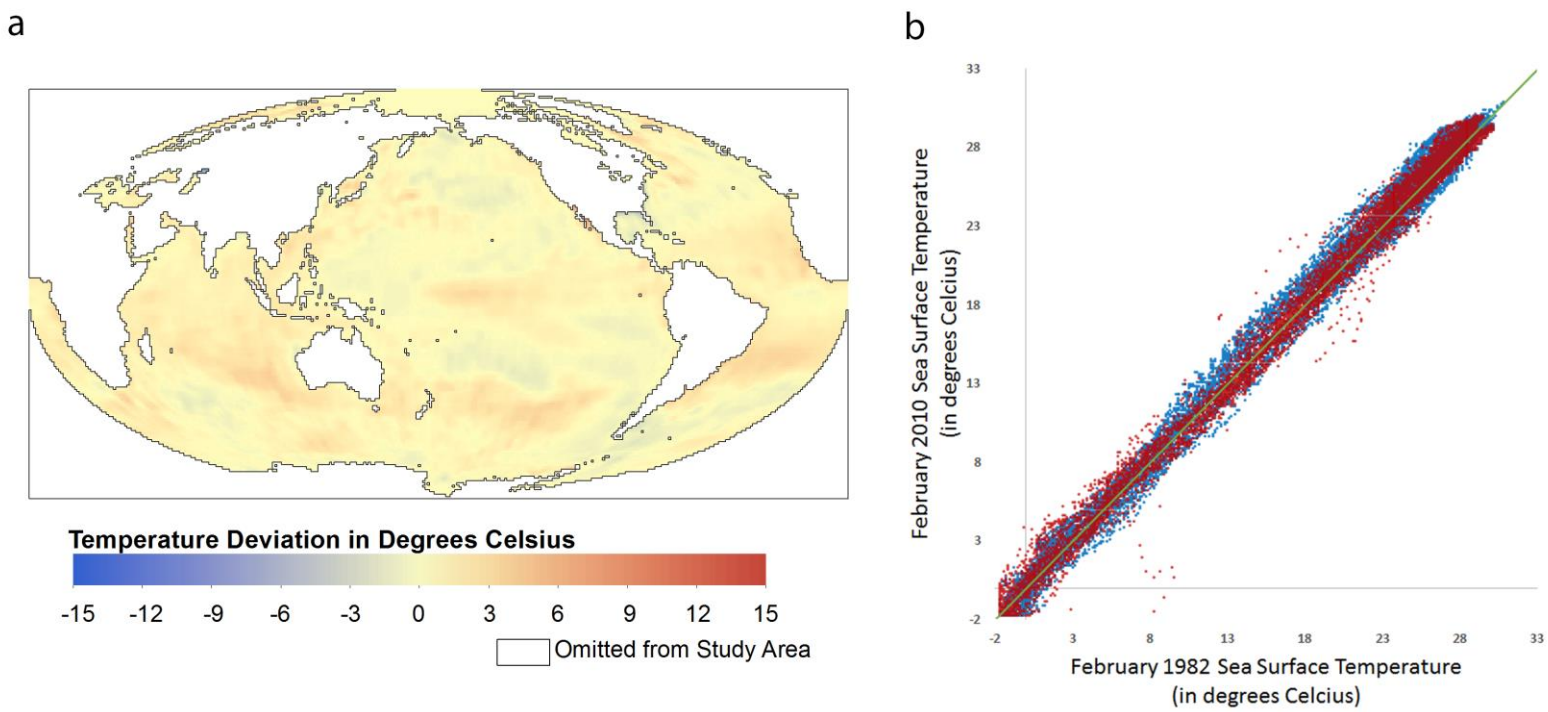

Figure 3. Map and scatterplot of temperature deviations for February 2010 minus February 1982. Each pixel on the map is represented as a point on the scatter plot to the right, with pixels in the North Stratum displayed in red and pixels in the South Stratum displayed in blue. The Y=X line in green shows where temperature values are the same for both time points.

2.2 Mean Absolute Deviation

2.2.1 Equations

Table 1 provides notation for this study. Equation 1 expresses deviation as the values of Y minus the values of X. SIGN indicates either a positive or negative value. Equation 2 reveals the direction of quantity component. If the mean value of Y is greater than the mean value of X then sgn is positive. If the mean value of Y is less than the mean value of X then sgn is negative. Equation 3 computes the Quantity component, which is equal to the absolute difference between mean X and mean Y. Equation 4 computes the Stratum Component, which compares absolute deviations across strata. If at least one $Dsg_{rc} > 0$ in one stratum and at least one other $Dsg_{rc} < 0$ in another stratum, then $SMAD > 0$; otherwise $SMAD = 0$. Stratum and Quantity components are constant across grains. Equation 5 computes Element component at grain g. Data at the original resolution are expressed as grain of 1. If $Dsg_{rc} > 0$ in one pixel and $Dsg_{rc} < 0$ in another pixel, then $EMAD_g > 0$; otherwise $EMAD_g = 0$. The relation of these components is expressed as $QMAD \leq SMAD \leq EMAD_g$.

Table 1. Mathematical notation for MAD with Stratification at Multiple Resolutions.

Notation	Meaning
Xsg_{rc}	the X value in stratum s at grain g for the element in row r and column c.
Ysg_{rc}	the Y value in stratum s at grain g for the element in row r and column c.
Dsg_{rc}	the Ysg_{rc} minus Xsg_{rc} deviation in stratum s at grain g for the element in for row r and column c.
g	grain, meaning the number of times greater the length of a coarse pixel side is compared to the length of the original resolution pixel side.
S	the number of strata.
N	the number of elements at the finest grain.
Rg	the number of rows at grain g.
Cg	the number of columns at grain g.

Dsg_{rc} = Ysg_{rc} - Xsg_{rc} (Equation 1)

$sgn = SIGN(\sum_{r=1}^{R1} \sum_{c=1}^{C1} Ds1_{rc})$ (Equation 2)

Quantity Component = $QMAD = \frac{\left|\sum_{s=1}^{s} \sum_{r=1}^{Rg} \sum_{c=1}^{Cg} Ds1_{rc}\right|}{N}$ (Equation 3)

Stratum Component = $SMAD = \frac{\sum_{s=1}^{s}\left|\sum_{r=1}^{Rg} \sum_{c=1}^{Cg} Ds1_{rc}\right|}{N} - QMAD$ (Equation 4)

Element Component at grain g = $EMAD_g = \frac{\sum_{s=1}^{s} \sum_{r=1}^{Rg} \sum_{c=1}^{Cg} |Dsg_{rc}|}{N} - SMAD - QMAD$

(Equation 5)

2.2.2 Multiple Resolutions

Total Deviation can be a function of spatial resolution. The inclusion of multiple resolution analysis influences only the element component. Equation 6 coarsens the spatial resolution of a deviation map by summing the deviations of pixels at the original resolution into coarser pixels, where each coarser pixel coincides with g columns and g rows of pixels at the original resolution. Equation 7 computes the number of rows at resolution g and equation 8 computes the number of columns at resolution g, where CEILING is a function that rounds up to the next highest integer.

$$Dsg_{rc} = \sum_{i=g(r-1)+1}^{gr} \left[\sum_{j=g(c-1)}^{gc} (Y_{ij} - X_{ij}) \right] \quad \text{(Equation 6)}$$

$$Rg = CEILING({}^{R1}/_{g}) \quad \text{(Equation 7)}$$

$$Cg = CEILING({}^{C1}/_{g}) \quad \text{(Equation 8)}$$

We have produced software, available to both Mac OS and Windows users, that computes MAD and its components according to the algorithms presented in this paper. Figure 4 describes the process for programming applications. Input images are read and an array of deviations is created. The programs iterative process begins by looping through strata, ignoring all masked pixels outside of the current stratum. The program then seeks possible coarser resolutions, using a coarsening factor defined by the user, until all elements within a stratum are encompassed into a single deviation. Coarsening takes place within the stratum loop. Elements in different strata are not aggregated when coarsening for multiple resolution analysis. At the end of these processes the program outputs total, stratum, quantity, and element level MAD. Output graphics allow users to understand the

distribution of deviations in relation to the Y=X line with a scatter plot, as well as the composition of different components relative to total MAD with a stacked bar graph.

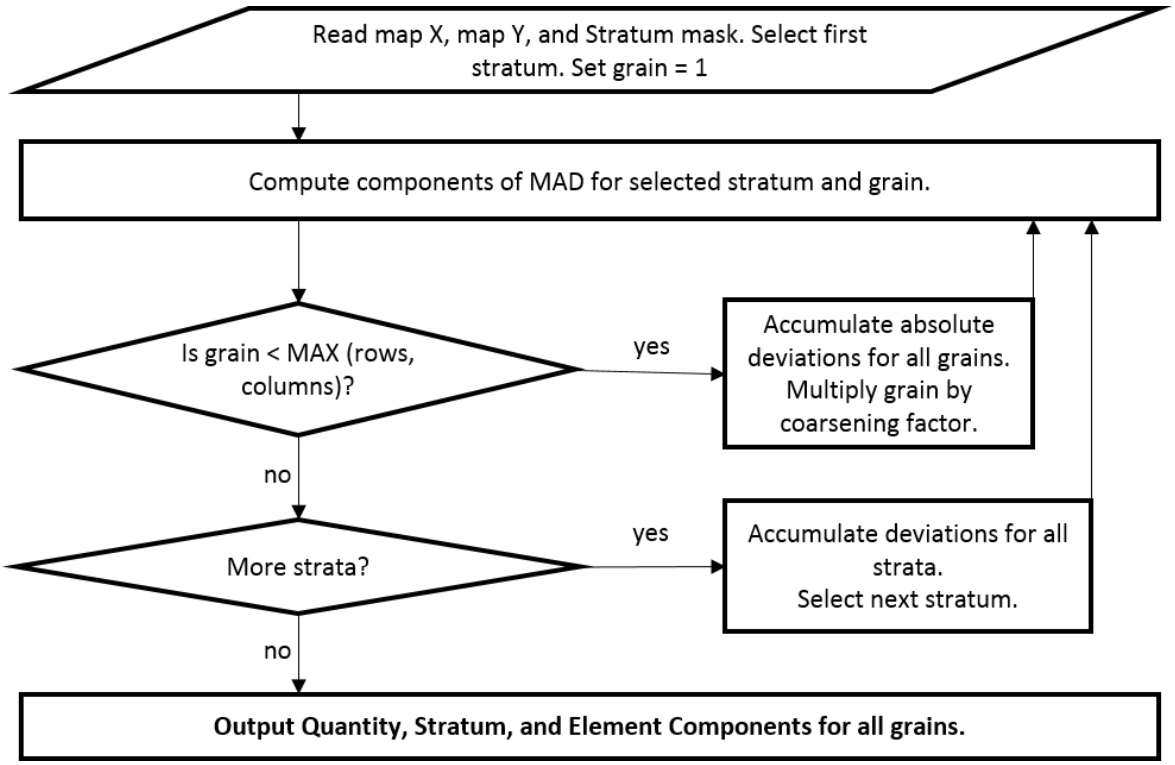

Figure 4. Flow Diagram showing the logic for programming applications to automate map comparison using the Mean Absolute Deviation.

3 Results

Figure 5 shows MAD at various grains using the February to August 1982 example. Total MAD is 3.7 degrees Celsius at grain 1, which is identified by the height of the entire bar. From February 1982 to August 1982, there is a quantity component of 0.07 degrees Celsius. *Sgn* is negative, indicating that the mean value of February 1982 is greater than the mean value of August 1982. Quantity component is a small share of total MAD

for all grains. There is a stratum component of 3.5 degrees, which makes up an overwhelming majority of total MAD at grain 1. Quantity and Stratum components remain constant across grains. There is an element component of 0.1 degrees Celsius at grain 1. As grain increases, we see lower values for element component.

Figure 6 shows MAD at various grains using the February 1982 to February 2010 example. Total MAD, indicated by the height of the entire bar, is 0.7 degrees Celsius at grain 1. There is a quantity component of 0.4 degrees Celsius. Half of total MAD at grain 1 is the quantity component. The mean value of February 2010 is greater, as indicated by a positive sign. There is no stratum component in the February 1982 to 2010 example because both strata have an overall positive deviation. Element component is 0.4 degrees Celsius at grain 1, which is half of total MAD. The coarsening map series illustrates the shrinking element component when red and blue fine-resolution pixels aggregate to yellow pixels in larger grains.

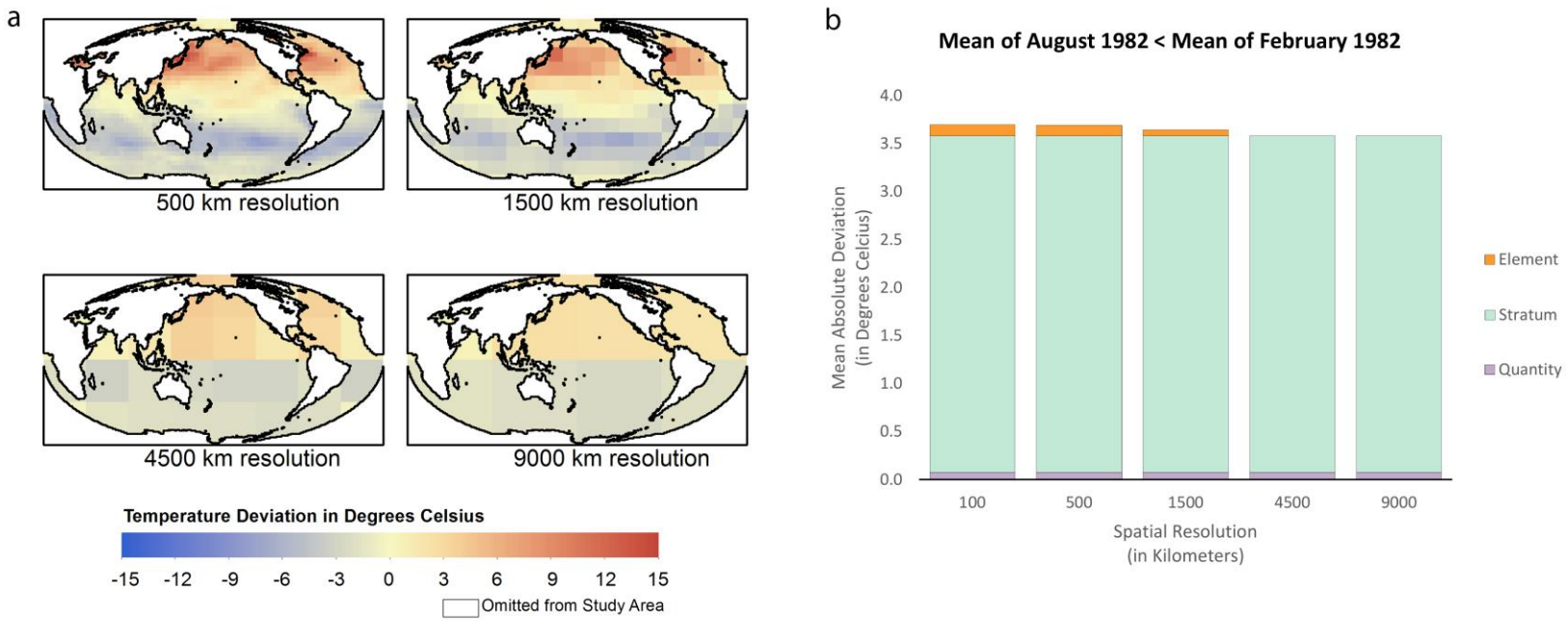

Figure 5. (a) The deviation map from February 1982 to August 1982 at four coarser resolutions. (b) Components if MAD at grains 5, 15, 45, and 90. The stacked bar graph shows the values of Quantity, Stratum, and Element components, adding to total Mean Absolute Deviation across multiple resolutions.

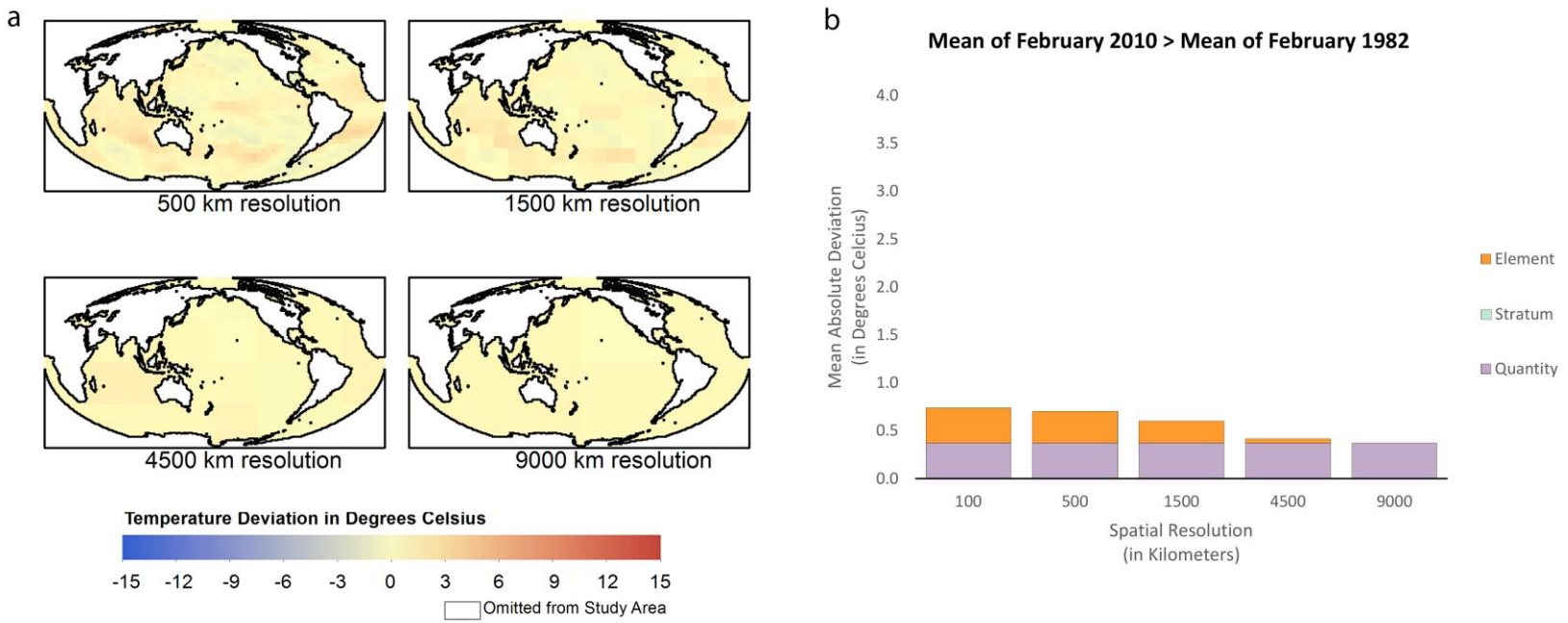

Figure 6. (a) The deviation map from February 1982 to February 2010 at four coarser resolutions. (b) Components if MAD at grains 5, 15, 45, and 90. The stacked bar graph shows the values of Quantity, Stratum, and Element components, adding to total Mean Absolute Deviation across multiple resolutions.

4 Discussion

4.1 Interpretation

MAD provides a mathematical way to characterize common types of deviations. The differences in Sea Surface Temperatures observed from February 1982 to August 1982 can be explained by earth's seasons. The northern hemisphere warms as it changes from winter to summer, and the southern hemisphere cools as it changes from summer to winter. The differences in Sea Surface Temperatures observed from February 1982 to February 2010 can be explained by global warming.

4.2 Components of Mean Absolute Deviation

The quantity component is very small for the seasonal comparison, indicating that global sea surface temperature is approximately the same in February 1982 as it is in August 1982. The stratum component, which is the largest component in the February 1982 to August 1982 example, tells us that there are positive temperature deviations in one stratum and negative temperature deviations in the other. This information confirms that change is due to alternating seasons, and that warming in the Northern stratum is matched by cooling in the Southern stratum. The small element component tells us that positive and negative deviations within each stratum do not mix very much as pixels coarsen.

The overall mean value of February 2010 is found to be greater that the overall mean value of February 1982, confirming warming. Quantity component makes up a large portion of the deviation in this example, indicating that warming is the predominant force

of change. Both strata have overall warming as indicated by the absence of stratum component for the February 1982 to February 2010 example. Element component, which makes up a large share of total MAD at grain 1, tells us that there are both positive and negative deviations in the same stratum. Therefore, warming happens unevenly within each stratum.

4.3 Characteristics of MAD

MAD has properties that make the approach easy to interpret and adopt in a wide variety of applications. MAD has the same units as X and Y, therefore MAD is interpretable in terms of the phenomenon that X and Y portray. MAD is bounded only by a lower limit of zero, which occurs if and only if X equals Y for all pixels. MAD has no upper limit because X and Y frequently have no limit, in which case there is no limit to the sizes of the deviations. MAD never leads to division by zero because the denominator of MAD is the number of pixels at the original resolution, which is always positive. When MAD compares X to Y we will get the same results as when MAD compares Y to X. Symmetry is desirable when it is not clear which variable should be defined as X and which variable should be defined as Y. The influence of each deviation on MAD is proportional to the absolute value of the deviation, which is intuitive. Lastly, MAD can be separated by its interpretable components, and is able to distinguish quantity deviation from allocation deviation.

5 Conclusions

To summarize, total Mean Absolute Deviation is the average distance of points from the Y= X line in a scatter plot. Quantity component is the absolute difference in mean value between maps X and Y. Stratum component measures deviations across strata, and Element measures deviations within each stratum.

MAD is useful for GIS professionals to measure differences in any application where maps present the same continuous variable. MAD is also effective and advantageous in assessing model or sensor performance. Components of MAD can be useful even when data are not spatially explicit. We ask that scientists report MAD when comparing maps of the same continuous variable. We have facilitated this approach by distributing software, free of charge, at chapmanrebecca.com/mad.

Acknowledgements

The students in the class GIS and Accuracy Assessment at Clark University contributed to the development of programming logic and provided useful feedback. The 2014 Peter J. Condakes Fellowship funded Summer Research. Any opinions, findings, conclusions, or recommendations expressed in this article are those of the authors and do not necessarily reflect those of the funders. Clark Labs facilitated this work by creating the GIS software Idrisi®, and supplying data for this project.

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